

Topological Analysis of Plant Structure Using Persistent Homology for Genotype Classification*

Prasenjit Betal
Indian Institute of Technology
Kharagpur
Kharagpur, India
prasenjibetal.23@kgpian.iitkgp.ac.in

Pabitra Mitra
Indian Institute of Technology
Kharagpur
Kharagpur, India
pabitra@cse.iitkgp.ac.in

Arindam Dasgupta
Indian Institute of Technology
Kharagpur
Kharagpur, India
adgkpg@gmail.com

Chittaranjan Mandal
Indian Institute of Technology
Kharagpur
Kharagpur, India
chitta@iitkgp.ac.in

Prateek Gupta
Indian Institute of Technology
Kharagpur
Kharagpur, India
pg7729600@gmail.com

ABSTRACT

Accurate quantification of complex plant architecture is critical for genetic improvement, yet commonly used morphological descriptors capture only local shape statistics and are highly sensitive to noise, viewpoint variation, and occlusions. This work introduces a phenotyping pipeline that integrates Persistent Homology (PH) with conventional skeleton-based descriptors to robustly characterize the structural organization of *Brassica juncea* plants from side-view and top-view RGB imagery. Euclidean distance filtrations applied to binary masks produce Persistence Diagrams (PDs), from which ten numerical PH statistics and fixed-resolution Persistence Images (PIs) are extracted. These topological descriptors encode canopy complexity and internal void structures that are poorly represented by traditional metrics. The PH-derived features are combined with geometric measurements such as object size and skeleton connectivity parameters. Experiments across multiple machine-learning classifiers demonstrate that this integrated representation yields substantial improvements in genotype classification accuracy. The proposed framework provides a scalable and noise-resilient solution for high-throughput plant phenotyping, enabling more reliable automated trait analysis.

CCS CONCEPTS

• **Computing methodologies** → **Machine learning**; • **Applied computing** → *Computational biology*; *Life and medical sciences*; • **Mathematics of computing** → *Topological data analysis*.

KEYWORDS

Plant Phenotyping, Genotype Classification, Persistent Homology (PH), Topological Data Analysis (TDA), Morphological Features, RGB Imaging, Machine Learning, Structural Pattern Analysis

*Produces the permission block, and copyright information

ACM Reference Format:

Prasenjit Betal, Pabitra Mitra, Arindam Dasgupta, Chittaranjan Mandal, and Prateek Gupta. 2025. Topological Analysis of Plant Structure Using Persistent Homology for Genotype Classification. In *Indian Conference on Computer Vision, Graphics, and Image Processing (ICVGIP 2025)*, December 17–20, 2025, Mandi, India. ACM, New York, NY, USA, 8 pages. <https://doi.org/10.1145/3774521.3774623>

1 INTRODUCTION

Understanding plant architecture is essential for deciphering the complex genotype-to-phenotype relationships that drive growth, development, and stress resilience in crops. Morphological traits such as leaf angle, curvature, branching patterns, and symmetry derived from both object characterization parameters and skeleton-based descriptors, significantly influence key physiological processes, including light interception, transpiration efficiency, and yield potential [10] [28]. Therefore, accurate and efficient characterization of plant morphology is vital for both fundamental plant research and practical applications in crop improvement and breeding programs.

With the advent of high-throughput phenotyping platforms, RGB imaging has become a non-invasive and scalable method for capturing plant structural features across time. From these images, diverse shape and structural traits such as area, compactness, eccentricity, skeleton length, leaf insertion points, and tip angles can be extracted using tools like PlantCV [11] and custom computer vision pipelines [22] [20]. These morphological and skeleton-based descriptors offer valuable insights into plant architecture, especially when integrated with temporal modeling techniques such as deep learning.

However, despite the richness of these extracted features, conventional morphometric and geometric descriptors often suffer from notable limitations. These approaches typically depend on predefined shape assumptions and operate on fixed spatial scales, making them inherently sensitive to noise, occlusion, or deformation in plant images. For instance, skeleton path length or convex hull area can be significantly affected by image resolution, camera angle, or partial occlusion of plant parts, leading to unreliable or non-repeatable measurements. Moreover, such descriptors often overlook the global structural integrity of plant architecture or



repeated motifs that reflect complex growth patterns. Recent studies have highlighted these challenges, emphasizing the need for more robust, scale-invariant, and topologically aware methods to describe plant morphology [23][21] [9].

To address these challenges, the incorporation of a mathematical framework known as Topological Data Analysis (TDA), specifically Persistent Homology (PH), into plant morphological analysis is proposed. This framework studies the shape and structure of complex data by focusing on how data points connect and form patterns across multiple scales, without relying on traditional coordinate systems or strict statistical assumptions [4] [5] [33]. Instead of analyzing isolated points, TDA identifies global structural features such as clusters, loops, and voids that persist over varying spatial resolutions, allowing for the distinction of meaningful patterns from noise. This scale-invariant approach makes TDA a valuable method for plant phenotyping, where capturing the overall form and spatial relationships within biological structures is essential for accurate morphological characterization [23] [9].

Persistent Homology, a key technique within TDA, captures topological features such as connected components (0-dimensional), loops or holes (1-dimensional), and voids (2-dimensional cavities). These features emerge, disappear, and vary as the spatial resolution or filtration parameters of images change. This process, known as *filtration*, constructs a nested sequence of shapes from the raw data, allowing the tracking of the birth and death of these features across scales. The lifespan of each topological feature is encoded in *persistence diagrams* and *barcodes*, providing compact and highly informative summaries of the underlying structural patterns present in image-based or point cloud data [8] [4].

This research fills a gap by leveraging Persistent Homology to analyze topological signatures embedded in RGB-derived morphological and skeleton features of plant shoots. Specifically, shape contours and skeleton graphs are transformed into simplicial complexes and homology in dimensions 0 and 1 is computed to identify connected regions and loops—representing branching patterns, gaps, or repeated motifs in plant structure. These persistent features are vectorized (e.g., using persistence images or summary statistics) and used to enrich phenotypic descriptors beyond conventional geometry.

In contrast to classical metrics, these topological features are resilient to noise, invariant to scale and orientation, and capture global structural relationships, making them ideal for comparing genotypes or developmental stages. Integration of PH-derived topological vectors with machine learning models enables classification of genotypes, tracking of growth phases, or detection of subtle structural differences, providing a powerful new layer of plant phenotypic analysis.

Persistent Homology (PH) can be applied to both static and time-series plant imaging data, though its utility differs significantly across these scenarios. When applied to time-series data (e.g., daily images over several days), PH enables the tracking of topological feature evolution—such as the emergence or disappearance of loops and connected components—offering insights into growth dynamics, developmental patterns, or stress responses. This temporal perspective enhances phenotypic characterization by capturing how plant structure changes over time. In contrast, when only a

single time-point image is available, PH still provides valuable descriptors of static morphology, including canopy complexity and internal voids, but lacks the ability to model developmental trends. Therefore, while PH is useful in both contexts, its full potential is best realized with multi-day time-series data that reflect the dynamics of plant architecture.

Beyond methodological contributions, PH-based descriptors have growing relevance in several emerging plant architecture applications. For mechanical harvesting, quantifying canopy compactness, branching uniformity, and global plant structure is essential for identifying crop ideotypes compatible with automation and robotics [12, 19]. In space-limited and vertical farming systems, topology-aware measures of canopy spread, density, and internal void distribution help optimize architectural traits for efficient light use and high-density growth [1, 16]. Persistent Homology is also valuable for stress phenotyping, as topological deviations can reflect structural changes caused by drought, heat, or nutrient stress, enabling robust detection of early stress responses [24, 25, 28]. Thus, the topological framework introduced here contributes meaningfully to modern goals in crop improvement, controlled-environment agriculture, and the development of stress-resilient plant ideotypes.

Random Forest: Random Forest is an ensemble learning algorithm [3] that builds multiple decision trees during training and combines their results to make a final prediction. Each tree is trained on a random subset of the data, which helps reduce overfitting and improves generalization. It's particularly useful when the dataset has a lot of features or noise. In this study, Random Forest was chosen for its robustness and ability to handle both simple and complex patterns effectively.

Multilayer Perceptron (MLP): MLP is a type of neural network [26] made up of multiple layers of connected neurons. It learns to map input features to the correct output class by adjusting the weights of connections through a process called backpropagation. MLPs are capable of capturing non-linear relationships in the data, making them suitable for complex classification problems. In our experiments, the MLP was configured with hidden layers and ReLU activation functions to efficiently learn from the combined skeleton and persistent homology features.

Support Vector Machine (SVM): Support Vector Machine [7] is a supervised learning method used for classification tasks. It works by finding the best possible boundary—called a hyperplane—that separates data points from different classes. SVM is especially effective when dealing with high-dimensional data and performs well when there's a clear gap between the classes. In this work, we used the radial basis function (RBF) kernel, which allows the model to handle more complex, non-linear patterns in the data.

2 METHODOLOGY

Algorithm 1 takes as input an RGB image α_i and performs a sequence of preprocessing and feature extraction steps. The image first undergoes preprocessing operations such as cropping and color adjustment, after which plant segmentation is applied to isolate the foreground plant structure, resulting in a binary mask M_i . A skeletonization process is then used to extract the medial axis of the plant, denoted as S_i , which effectively captures the overall branching pattern and topology.

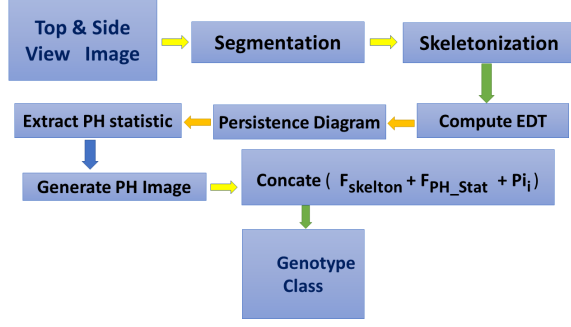


Figure 1: flow Diagram

Input: Image set $P = \{\alpha_1, \alpha_2, \dots, \alpha_n\}$ (RGB side/top view)

Input: Genotype labels $G = \{g_1, g_2, \dots, g_n\}$

Output: Predicted genotype classes $Y = \{y_1, y_2, \dots, y_n\}$

foreach $\alpha_i \in P$ **do**

```

  I ← preprocess( $\alpha_i$ ) ;      // Crop, rotate, color
  convert
   $M_i$  ← segmentPlant(I) ;    // Thresholding +
  morphological ops
   $S_i$  ← skeletonize( $M_i$ ) ;    // Generate skeleton
   $S'_i$  ← pruneSkeleton( $S_i$ , threshold) ; // Remove small
  spurs
   $F_{skel}$  ← extractSkeletonFeatures( $S'_i$ ) ;
   $D_i$  ← computeEDT( $M_i$ ) ;    // Euclidean Distance
  Transform
   $PD_i$  ← computePersistenceDiagram( $D_i$ ) ;
   $F_{PHstat}$  ← extractPHStatistics( $PD_i$ ) ;
   $PI_i$  ← generatePersistenceImage( $PD_i$ ) ;
   $F_i$  ← concatenate( $F_{skel}$ ,  $F_{PHstat}$ ,  $PI_i$ ) ;
  Append  $F_i$  to feature set  $F$  ;
  Append  $g_i$  to label set  $G$  ;

```

end

```

model ← trainClassifier(F, G) ; // Random Forest, SVM,
etc.

```

```

 $Y_{pred}$  ← model.predict(F) ;

```

return Y_{pred} ;

Algorithm 1: Genotype Classification with Skeleton and Persistent Homology Features

To eliminate noise and irrelevant fine structures, small spurs are pruned from the skeleton, producing a refined skeleton S'_i . From this cleaned representation, a set of N skeleton-based features F_{skel} is extracted. These features encode important morphological characteristics such as the number of branches, total skeleton length, branching angles, and the distribution of endpoints—attributes that are critical for differentiating plant genotypes based on their structural configurations.

The dataset spans 10 distinct *Brassica juncea* (Indian mustard) genotypes—namely RE-2-2, RE-8, RE-13, RE-14, RE-35-4, CN-101849, CN-105306, CN-105308, CN-101846 and CN-112920. For each genotype, plants were imaged from both side and top views, capturing

complementary morphological characteristics and structural patterns relevant to genotype-specific growth behavior. All images were acquired under consistent conditions, with a maintained resolution of [insert resolution, e.g., 1024×768 pixels], ensuring sufficient detail for downstream topological and geometric feature extraction. The dataset is balanced across genotypes and provides a robust foundation for phenotypic analysis and genotype classification.



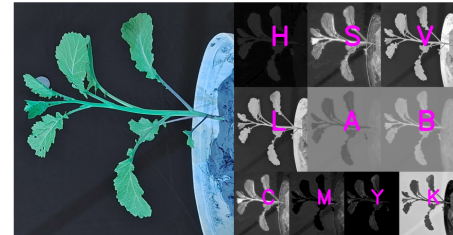
Figure 2: Image (CN-101846)

To ensure robust extraction of structural features from RGB plant images, a multi-stage preprocessing and segmentation pipeline was implemented using the PlantCV library [11] in combination with additional custom procedures to handle specific analysis steps.

The pipeline begins with colour-space transformations to enhance the visual contrast between plant tissue and background. Different colour representations offer distinct advantages depending on the imaging conditions and phenotyping objectives.

The RGB (Red, Green, Blue) colour space is an additive model commonly used in digital imaging; however, it is not perceptually uniform and often performs poorly in segmentation tasks under varying illumination. The HSV (Hue, Saturation, Value) colour space provides a more intuitive decomposition—where hue denotes colour type, saturation reflects purity, and value encodes brightness—making it useful for isolating specific colour ranges such as green foliage or yellowing stress indicators.

The CIELAB ($L^*a^*b^*$) colour space, illustrated in Figure 3, separates luminance (L^*) from two chromatic channels: a^* (green–magenta) and b^* (blue–yellow). Owing to its perceptual uniformity and robustness to lighting variation, LAB is widely used in plant image analysis for improving contrast between vegetation and non-vegetation regions. In contrast, CMYK (Cyan, Magenta, Yellow, Black) is a subtractive model primarily used in printing and is not typically employed in phenotyping pipelines.

Figure 3: CIELAB ($L^*a^*b^*$) colour space

To isolate the plant structure from the background, binary thresholding was applied to the a^* channel of the LAB colour space. Vegetation pixels exhibit distinctly lower a^* values than soil and pot surfaces, enabling effective separation in this chromatic dimension Figure 4. The resulting grayscale image was converted into a binary mask Figure 5 by assigning each pixel to either plant or background using an experimentally determined threshold that accommodates variation in lighting and contrast. The object type was specified as “light” to retain plant regions as the foreground. This segmentation step provides a clean and reliable mask that supports subsequent morphological processing and structural feature extraction.

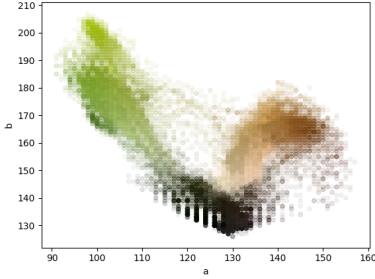


Figure 4: Scatter Diagram



Figure 5: Binary Mask

To localize the analysis and reduce background interference, a rectangular region of interest (ROI) was defined for each image, focusing processing on the central area where the plant is typically located in Figure 5. This targeted selection helped to exclude distracting elements such as pot rims or image borders etc. The binary mask was then filtered to include only pixels within the ROI.



Figure 6: Region of Interest (ROI)

Euclidean Distance Transform (EDT). Given a binary segmentation mask M_i , where $M_i(x, y) = 1$ for foreground (plant) pixels and 0 otherwise, we compute the Euclidean Distance Transform (EDT) to obtain a grayscale image D_i . Each pixel in D_i is assigned the shortest distance to the nearest background pixel, formally computed as:

$$D_i(x, y) = \min_{(x', y') \in \text{background}} \sqrt{(x - x')^2 + (y - y')^2}.$$

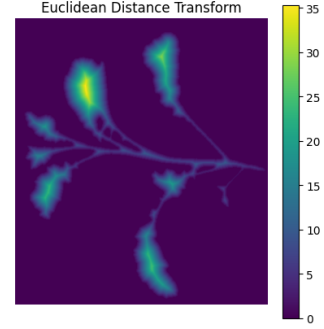


Figure 7: Euclidean Distance transform

This transformation provides a “landscape” that highlights the thickness of plant structures—pixels farther from the background receive higher values, effectively encoding the internal density of the plant.

To extract topological features from the plant’s shape, we begin by computing a Euclidean Distance Transform (EDT) of the plant mask M_i , producing a grayscale image D_i . This distance map encodes how deep each pixel is within the plant structure — interior pixels have higher values, while boundary pixels are closer to zero.

From D_i , we construct a topological filtration using a sequence of superlevel sets:

$$F_t = \{(x, y) \mid D_i(x, y) \geq t\}$$

where t varies from the maximum to the minimum pixel value in D_i . As t decreases, more of the plant structure is included in the superlevel set F_t , gradually revealing the entire shape.

We treat the image domain as a 2D cubical complex, where each pixel is a 2-cell (square), and the filtration creates a nested sequence of topological spaces.

$PD_i \leftarrow \text{computePersistenceDiagram}(D_i)$

Once we have the Euclidean Distance Transform D_i , the next step is to extract a **Persistence Diagram** PD_i using topological data analysis (TDA). This diagram captures the appearance and disappearance of topological features—such as connected components and loops—during a filtration process based on pixel distance values.

- **Filtration:** A *superlevel set filtration* is applied to D_i , where pixels with higher values (typically interior plant regions) are included first, and lower values (near edges) later.

- A **cubical complex** representation—well-suited for grid-based image data—is used to track topological changes throughout this filtration.
- As the threshold decreases:
 - New **connected components** (0-dimensional features, H_0) are born.
 - Existing components may merge, leading to the **death** of one.
 - **Loops or holes** (1-dimensional features, H_1) can form and eventually get filled.

Output: Persistence Diagram PD_i : The result is a multiset of birth-death pairs (b_j, d_j) , where:

- b_j : filtration value when a topological feature appears,
- d_j : filtration value when it disappears.

Each point in the persistence diagram corresponds to a single topological feature. Features with a large persistence $d_j - b_j$ (i.e., far from the diagonal) are typically significant. In contrast, short-lived features near the diagonal may represent noise.

- Captures **global structural traits** like symmetry, repeated motifs, and internal voids.
- Robust to **noise, deformation, and scale**, offering complementary insights to conventional shape descriptors.
- Enables **quantification of complex morphology** in a principled, scale-invariant way.

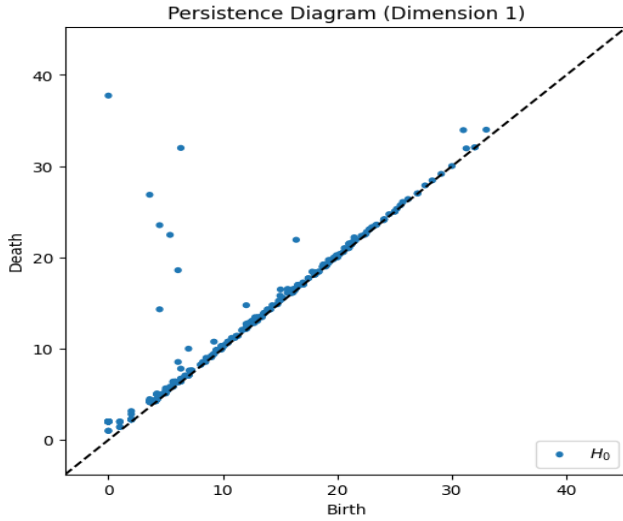


Figure 8: Persistence Diagram

Figure 8 shows the persistence diagram, where each point represents a topological feature with coordinates corresponding to its birth and death scales. Features that persist longer—that is, points located farther from the diagonal—indicate more significant structural components extracted from the Euclidean Distance Transform of the plant image.

Topological Features and Summary Statistics

The following topological features were extracted from the persistence diagrams of connected components (H_0) and loops (H_1). For

the connected components (H_0), we computed: (1) the total number of connected components, denoted as $|PD_i^{H_0}|$; (2) the maximum persistence, calculated as $\max_j (d_j - b_j)$; and (3) the mean persistence, given by $\frac{1}{|PD_i^{H_0}|} \sum_j (d_j - b_j)$.

For the loops or holes (H_1), the following features were extracted:

- (1) **Total number of loops:** $|PD_i^{H_1}|$.
- (2) **Maximum persistence:**

$$\max_j (d_j - b_j).$$

- (3) **Mean persistence:**

$$\frac{1}{|PD_i^{H_1}|} \sum_j (d_j - b_j).$$

- (4) **Total persistence:**

$$\sum_j (d_j - b_j).$$

- (5) **Loop density:** number of loops per unit area A ,

$$\frac{|PD_i^{H_1}|}{A}.$$

- (6) **Loop area:** weighted persistence measure,

$$\sum_j (d_j - b_j)^2.$$

- (7) **Loop persistence ratio:** dominance of the most persistent loop,

$$\frac{H1_MaxPersistence}{H1_MeanPersistence + \epsilon},$$

where ϵ is a small constant to avoid division by zero.

Persistence Image Generation:

To provide a fixed-dimensional, differentiable topological descriptor suitable for machine learning, we convert each persistence diagram into a persistence image (PI). First, all (b, d) pairs are transformed into (b, p) , where $p = d - b$, emphasizing persistence. This new coordinate space is discretized into a uniform grid (typically 30×30 bins). For each transformed point, a Gaussian kernel is centered at (b, p) and spread across the grid. The pixel value at location (x, y) is computed by summing:

$$I(x, y) = \sum_{(b, p) \in PD} \exp\left(-\frac{(x - b)^2 + (y - p)^2}{2\sigma^2}\right),$$

where σ controls the spread of each point's contribution. The resulting 30×30 matrix is flattened into a 900-dimensional vector $PI_i \in \mathbb{R}^{900}$. This persistence image preserves spatial information of topological features in a format suitable for standard classifiers.

Figure 9 presents the resulting persistence image. After transforming the (b, d) pairs into (b, p) coordinates, the birth-persistence space is discretized into a uniform 30×30 grid. A Gaussian kernel is then centered at each (b, p) point, allowing its influence to spread smoothly across neighboring grid cells. This converts the sparse set of topological features from the persistence diagram into a dense heatmap representation. High-intensity regions in the persistence

image indicate strongly persistent and structurally important features, whereas low-intensity regions arise from short-lived features located near the diagonal in the diagram. The resulting 30×30 matrix is finally flattened into a 900-dimensional vector, providing a fixed-size descriptor that can be directly used as input to machine-learning models.

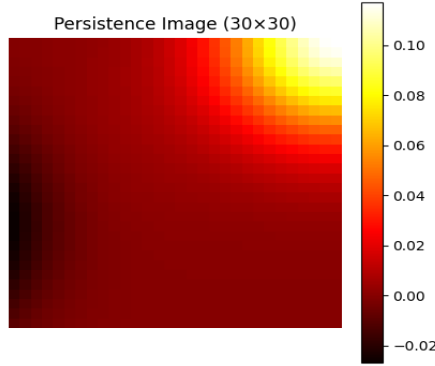


Figure 9: Persistence Image

3 EXPERIMENTAL RESULT

The skeleton-based structural descriptors (F_{skel}) encompass geometric and shape-related measurements. The descriptions of these features are as follows[17]:

Convex Hull Area. The convex hull area refers to the area of the smallest convex polygon that fully encloses the plant or object. It is a useful descriptor for understanding the outer boundary and spatial spread of a shape. It is often used in comparison with the actual object area for computing solidity and compactness.

Solidity. Solidity measures how completely the shape fills its convex hull. It is calculated as:

$$\text{Solidity} = \frac{\text{Object Area}}{\text{Convex Hull Area}}$$

A value close to 1 indicates a compact and convex shape, while lower values reflect concavities or holes.

Segment Insertion Angle. This feature quantifies the angle at which lateral branches or segments are connected to the main stem. It captures architectural traits such as the spread of leaves or branches, which are essential in characterizing plant phenotypes.

Segment Tangent Angle. Segment Tangent angle represents the orientation of the skeleton at each point along its path. It is computed as the angle between the local tangent vector and a reference axis (typically horizontal), offering insight into the directional flow of the shape.

Segment Angle. Segment angle measures the angular deviation between consecutive segments of the skeleton. It reflects how abruptly or smoothly the object changes direction, which is especially useful in assessing structural complexity.

Segment Curvature Angle. This feature quantifies the bending or curvature of a segment along the skeleton. It is generally defined as the rate of change of the tangent angle with respect to arc length, and is indicative of wavy or coiled structural patterns.

Branch Point Count. Branch point count refers to the number of junctions in the skeleton where branches split into multiple paths. It reflects the complexity and connectivity of the shape and is particularly relevant in biological structures like plants.

Tip Point Count. Tip point count represents the number of end-points or terminations in the skeleton. It is useful for estimating the number of leaves or extremities and provides insights into the growth patterns of a structure.

Compactness. Compactness is a shape descriptor that indicates how efficiently space is utilized by the structure. It is defined as:

$$\text{Compactness} = \frac{\text{Object Area}}{\text{Convex Hull Area}}$$

Higher compactness values indicate more filled-in shapes, while lower values suggest branching or sparse configurations.

Convex Hull. This feature measures the total length of the boundary of the convex hull. It simplifies the shape to its outermost extent and is often used alongside convex hull area to assess regularity and spread.

Contour Area. Contour area is the area enclosed by the outer boundary of the object. It includes all internal pixels of the object but excludes any holes. It is a basic yet critical parameter for further morphological analysis.

Euclidean Length. Euclidean length is the straight-line distance between two endpoints of a skeleton segment. It is calculated using the formula:

$$\text{Length} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

This measure gives an estimate of the direct extent of a segment without accounting for curvature.

Segment Path Length. Path length represents the actual length of a segment traced along its curve or skeleton. It is calculated by summing the distances between consecutive skeleton points. This feature is more accurate than Euclidean length for curved or meandering structures.

Eccentricity. Eccentricity quantifies how elongated an object is based on the best-fitting ellipse. It is defined as:

$$\text{Eccentricity} = \sqrt{1 - \frac{b^2}{a^2}}$$

where a and b are the lengths of the major and minor axes, respectively. A value of 0 indicates a circular shape, while values approaching 1 reflect high elongation.

The second component, persistent homology summary statistics (F_{PHstat}), captures topological information derived from persistence diagrams. Finally, the third component is the persistence image (PI_i), a 900-dimensional vector obtained by flattening a 2D representation of the persistence diagram. This image encodes the spatial distribution and significance of topological features in a form suitable for machine learning. The combined vector $F_i \in \mathbb{R}^{921}$ thus provides a

comprehensive morphological and topological characterization of each plant image.

Feature Concatenation: Once the individual feature components have been extracted for each plant image α_i , they are combined into a single high-dimensional feature vector F_i . This vector integrates three types of features.

All feature vectors are compiled into a dataset:

$$F = \{F_1, F_2, \dots, F_n\} \in \mathbb{R}^{n \times 921}$$

where n is the number of plant images.

Each vector F_i is paired with a genotype label g_i , forming the supervised label set:

$$G = \{g_1, g_2, \dots, g_n\}$$

Together, the dataset (F, G) serves as the input for downstream statistical and machine learning models. This integrated feature representation enables robust and interpretable classification of plant genotypes, especially where morphological variations are subtle or distributed across spatial scales.

Correlation Heatmap Analysis. The correlation heatmap Figure 10 illustrates the relationships among the skeleton-based structural descriptors, revealing feature dependencies and potential redundancies. Strong positive correlations are observed between size-related metrics such as Convex Hull Area and Contour Area, reflecting consistent measurement of overall plant size. Angular features, including statistics of Insertion Angle, Tangent Angle, and Segment Angle, show moderate correlations, indicating coordinated variations in branch orientations and curvature. Length-based features (mean and standard deviation of Euclidean Length and Path Length) correlate with branching complexity metrics such as Branch Point and Tip Point Counts, emphasizing their combined role in capturing structural complexity. This analysis aids in informed feature selection and improves interpretability of genotype classification models.

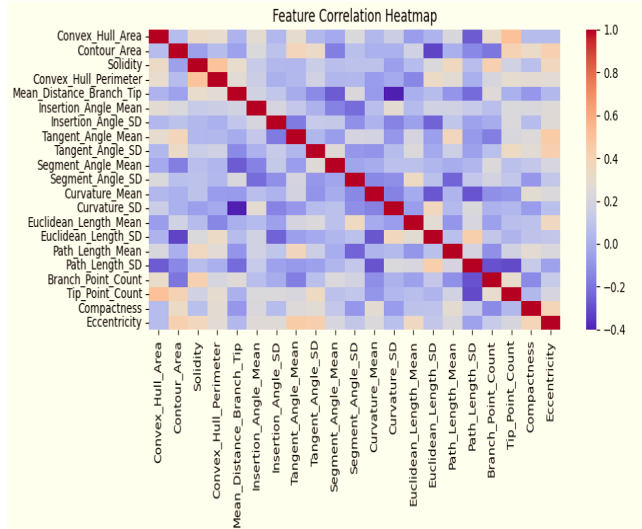


Figure 10: Correlation matrix

UMAP-Based Dimensionality Reduction and Analysis. Figure 11 was employed for dimensionality reduction, utilizing a comprehensive set of phenotypic descriptors derived from shape, skeleton, and structural features. Compared to other nonlinear methods, UMAP demonstrated superior cluster compactness and better preservation of the global data structure. As illustrated in Figure 9, the UMAP visualization distinctly separates the ten *Brassica juncea* genotypes into well-defined clusters.

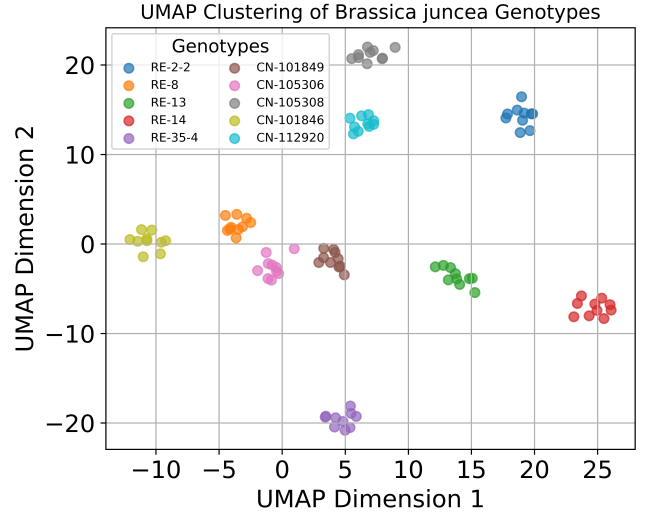


Figure 11: UMAP-based clustering of *Brassica juncea* genotypes.

Table 1: Comparison of classifier performance using different feature sets

Classifier	Feature Set	Accuracy (%)	F1-score (%)
SVM	Skeleton only	87.4	86.6
SVM	Skeleton + PH	93.7	93.1
Random Forest	Skeleton only	88.5	87.9
Random Forest	Skeleton + PH	94.2	93.6
MLP	Skeleton + PH	92.5	91.1

The experimental results demonstrate that the integration of persistent homology (PH) features with conventional skeleton-based descriptors significantly enhances genotype classification performance across all classifiers. Skeleton-only features, while informative of plant structure, tend to capture local branching patterns and length distributions. However, they often miss global spatial arrangements and voids that are critical for distinguishing complex canopy architectures. PH features, derived from topological data analysis, provide a robust representation of these global patterns by quantifying loops, connectivity, and voids across multiple spatial scales. When combined, the two feature sets offer complementary perspectives on plant morphology—skeleton features encode fine structural traits, while PH captures scale-invariant topological complexity. This fusion results in a richer, more discriminative feature

space, which explains the marked improvement in accuracy and F1-score observed in all classifiers, particularly Random Forest and SVM.

4 CONCLUSION

While the proposed phenotyping framework achieves strong genotype classification performance and demonstrates robust structural characterization through the integration of persistent homology and skeleton-based descriptors, several limitations remain. The dataset used in this study is relatively constrained, consisting of a fixed set of *Brassica juncea* genotypes captured under controlled imaging conditions. This restricted diversity may limit the generalizability of the model to field environments, where plant architecture exhibits greater variability due to fluctuating illumination, background clutter, and stress conditions.

Future work will focus on expanding the dataset to include a broader spectrum of genotypes, environmental settings, and acquisition setups. Incorporating temporal image sequences will enable the analysis of dynamic growth patterns, canopy development, and stress-induced architectural changes. Additionally, extending the pipeline to multimodal data such as depth maps, hyperspectral measurements, or 3D reconstructions could further enhance the discriminative power of the topological descriptors. These advancements will help establish a more scalable, noise-resilient, and biologically interpretable phenotyping framework suitable for high-throughput field applications.

REFERENCES

- [1] K. Benke and B. Tomkins. 2017. Future food-production systems: vertical farming and controlled-environment agriculture. *Sustainability: Science, Practice and Policy* 13, 1 (2017), 13–26.
- [2] L. T. Berner, T. H. White, J. P. Anderson, and J. T. Aber. 2022. Topological data analysis reveals plant root architecture variation. *New Phytologist* 234, 5 (2022), 1796–1808.
- [3] L. Breiman. 2001. Random forests. *Machine Learning* 45, 1 (2001), 5–32.
- [4] G. Carlsson. 2009. Topology and data. *Bull. Amer. Math. Soc.* 46, 2 (2009), 255–308.
- [5] F. Chazal and B. Michel. 2017. An introduction to topological data analysis: fundamental and practical aspects for data scientists. *arXiv preprint arXiv:1710.04019* (2017).
- [6] J. Chen, M. Zhang, and Y. Sun. 2023. Interpretable deep phenotyping using topological data analysis on crop 3D models. *Plant Methods* 19, 1 (2023).
- [7] C. Cortes and V. Vapnik. 1995. Support-vector networks. *Machine Learning* 20, 3 (1995), 273–297.
- [8] H. Edelsbrunner and J. Harer. 2008. Persistent homology—a survey. *Contemp. Math.* 453 (2008), 257–282.
- [9] Matthew Griffiths et al. 2023. Topological data analysis expands the genotype to phenotype map for 3D maize root system architecture. *Frontiers in Plant Science* (2023). <https://www.frontiersin.org/journals/plant-science/articles/10.3389/fpls.2023.1260005/full>
- [10] R. T. Furbank and M. Tester. 2020. Phenomics – technologies to relieve the phenotyping bottleneck. *Trends in Plant Science* 25, 3 (2020), 184–195.
- [11] M. A. Gehan, N. Fahlgren, A. Abbasi, J. C. Berry, S. T. Callen, L. Chavez, A. N. Doust, M. J. Feldman, K. B. Gilbert, J. G. Hodge, and et al. 2017. PlantCV v2: Image analysis software for high-throughput plant phenotyping. *PeerJ* 5 (2017), e4088.
- [12] G. Hammer, E. van Oosterom, R. McLean, and et al. 2019. Plant architecture and implications for crop design and harvestability. *Journal of Experimental Botany* 70, 18 (2019), 4481–4494.
- [13] K. He, X. Zhang, S. Ren, and J. Sun. 2020. Deep residual learning for image recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*.
- [14] Y. Huang, S. Li, and K. Chen. 2022. Topological data analysis for leaf vein network characterization. *Plant Physiology* 188, 2 (2022), 894–905.
- [15] E. Katifori, G. J. Szöllösi, and D. J. Siggia. 2021. Topological data analysis of plant vascular networks. *PNAS* 118, 9 (2021).
- [16] T. Kozai, G. Niu, and M. Takagaki. 2019. *Plant Factory: An Indoor Vertical Farming System for Efficient Quality Food Production*. Academic Press.
- [17] LemnaTec GmbH. n.d. LemnaTec High-Content Screening: LEMNATEC ANALYSIS PARAMETERS. *LemnaTec Website* (n.d.). <https://www.lemnatec.com/phenoaiexpert/> Accessed: 2025-08-11.
- [18] H. Li, Z. Huang, J. Zhao, and X. Li. 2021. Applying persistent homology to analyze leaf shapes for plant species classification. *Frontiers in Plant Science* 12 (2021).
- [19] L. Li, Q. Zhang, and D. Huang. 2014. A review of imaging techniques for plant phenotyping. *Sensors* 14, 11 (2014), 20078–20111.
- [20] Z. Liang, H. Zhou, and X. Wang. 2020. Skeleton-based plant phenotyping for leaf shape and growth analysis. *Computers and Electronics in Agriculture* 178 (2020).
- [21] Lukas Meyer, Andreas Gilson, Oliver Scholz, and Marc Stamminger. 2023. CherryPicker: Semantic Skeletonization and Topological Reconstruction of Cherry Trees. *arXiv preprint arXiv:2304.04708* (2023). <https://arxiv.org/abs/2304.04708>
- [22] M. Minervini, A. Abdelsamea, and S. Tsafaris. 2021. Image-based plant phenotyping with deep learning. *Trends in Plant Science* 26, 11 (2021), 1092–1104.
- [23] Haruhisa Oda, Mayuko Kida, Yoichi Nakata, and Hiroki Kurihara. 2024. Novel definition and quantitative analysis of branch structure with topological data analysis. *arXiv preprint arXiv:2402.07436* (2024). <https://arxiv.org/abs/2402.07436>
- [24] R. Pieruschka and H. Poorter. 2012. Phenotyping plant responses to environmental change. *Functional Plant Biology* 39 (2012), 839–850.
- [25] A. Raza, M. M. Siraj, and K. Lee. 2021. Machine learning for plant stress phenotyping. *Plant Science* 307 (2021).
- [26] D. E. Rumelhart, G. E. Hinton, and R. J. Williams. 1986. Learning internal representations by back-propagating errors. *Nature* 323, 6088 (1986), 533–536.
- [27] Y. Shi, X. Chen, and Z. Yu. 2023. Advanced imaging and machine learning for high-throughput plant phenotyping. *Trends in Plant Science* 28, 2 (2023).
- [28] A. Singh, K. Ganapathysubramanian, S. K. Singh, and A. Sarkar. 2021. Machine learning for high-throughput stress phenotyping in plants. *Trends in Plant Science* 26, 1 (2021), 88–100.
- [29] H. Tang, J. Zhou, and Y. Wu. 2021. A new framework for plant architecture analysis combining skeleton extraction and machine learning. *Plant Methods* 17, 1 (2021).
- [30] J. R. Ubbens, M. Cieslak, P. Prusinkiewicz, and I. Stavness. 2020. Latent space phenotyping: automatic image-based phenotyping for treatment studies. *Plant Methods* 16, 1 (2020), 1–16.
- [31] J. Wang, Y. Chen, and S. Jiang. 2021. Deep learning-based segmentation and skeleton extraction for plant root phenotyping. *Plant Methods* 17, 1 (2021).
- [32] Y. Wang, X. Zhang, Z. Zhao, and H. Xu. 2022. Explainable deep learning for plant phenotyping: A review. *Computers and Electronics in Agriculture* 195 (2022).
- [33] L. Wasserman. 2018. Topological data analysis. *Annual Review of Statistics and Its Application* 5 (2018), 501–532.
- [34] B. Xu, J. Tang, and H. Gao. 2020. A survey on explainable AI: Techniques and applications in plant science. *Artificial Intelligence Review* 53 (2020), 4973–4993.
- [35] L. Yang, M. Li, and S. Wu. 2023. Interpretable AI models for plant phenotyping using multi-modal imaging data. *Frontiers in Plant Science* 14 (2023).
- [36] X. Zhang, Y. Huang, and W. Zhang. 2021. Morphological and skeleton-based leaf phenotyping using deep learning. *Computers and Electronics in Agriculture* 187 (2021).
- [37] Q. Zhou, Y. Zhang, and H. Huang. 2022. Persistent homology-based machine learning for plant leaf shape analysis. *IEEE Transactions on Geoscience and Remote Sensing* 60 (2022).